

## Performance Testing

|                      |                                                             |
|----------------------|-------------------------------------------------------------|
| <b>Date</b>          | 03 July 2026                                                |
| <b>Team ID</b>       | xxxxxx                                                      |
| <b>Project Name</b>  | OptiCrop: Smart Agricultural Production Optimization Engine |
| <b>Maximum Marks</b> | 5 Marks                                                     |

### Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

#### Step 1: Testing Overview

| Field                      | Details                                 |
|----------------------------|-----------------------------------------|
| <b>Testing Tool Used</b>   | Flask Test Client, Browser Testing      |
| <b>Type of Testing</b>     | Functional Testing, Performance Testing |
| <b>Target Module / API</b> | Crop Prediction Module                  |
| <b>Test Environment</b>    | Windows 11, Python 3.11, Flask          |
| <b>Test Date</b>           | 03 July 2026                            |

#### Step 2: Test Scenarios

| S.No | Test Scenario / Description        | No. of Virtual Users | Duration (sec) | Expected Outcome           |
|------|------------------------------------|----------------------|----------------|----------------------------|
| 1    | Crop prediction using valid inputs | 1                    | 10             | Correct crop is displayed  |
| 2    | Invalid input values               | 1                    | 10             | Validation error shown     |
| 3    | Multiple prediction requests       | 10                   | 30             | System responds correctly  |
| 4    | Continuous prediction requests     | 20                   | 60             | Application remains stable |

### Step 3: Performance Test Results

| S.No | Metric              | Target Value | Actual Value      | Status | Remarks       |
|------|---------------------|--------------|-------------------|--------|---------------|
| 1    | Response Time (Avg) | < 2 sec      | <b>1.2 sec</b>    | Pass   | Fast response |
| 2    | Response Time (Max) | < 5 sec      | <b>2.3 sec</b>    | Pass   | Within limit  |
| 3    | Throughput          | High         | <b>25 req/sec</b> | Pass   | Stable        |
| 4    | Error Rate          | <1%          | <b>0%</b>         | Pass   | No errors     |
| 5    | CPU Utilization     | <80%         | <b>42%</b>        | Pass   | Normal        |
| 6    | Memory Utilization  | <80%         | <b>55%</b>        | Pass   | Stable        |

### Step 4: Observations & Analysis

#### Key Findings

- Crop prediction is generated quickly.
- The application remained stable during testing.

#### Bottlenecks Identified

- No major bottlenecks were observed.
- Response time increased slightly under multiple requests.

#### Optimization Steps Taken

- Optimized the trained ML model loading.
- Validated user inputs before prediction.

### Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

```
(venv) (base) PS C:\Users\saivi\Downloads\OptiCrop-main\OptiCrop-main> python app.py
[OptiCrop] Machine Learning model, scaler & encoder loaded successfully.
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://10.227.236.154:5000
Press CTRL+C to quit
* Restarting with stat
[OptiCrop] Machine Learning model, scaler & encoder loaded successfully.
* Debugger is active!
* Debugger PIN: 108-825-826
```